Smallest Difference

AlgoExpert · Medium · Arrays

The problem

- Pick one number from each array with the smallest absolute difference; return [a, b].

Intuition

- Sort both arrays, then walk them together, always advancing the smaller value to close the gap.

Brute force vs optimal

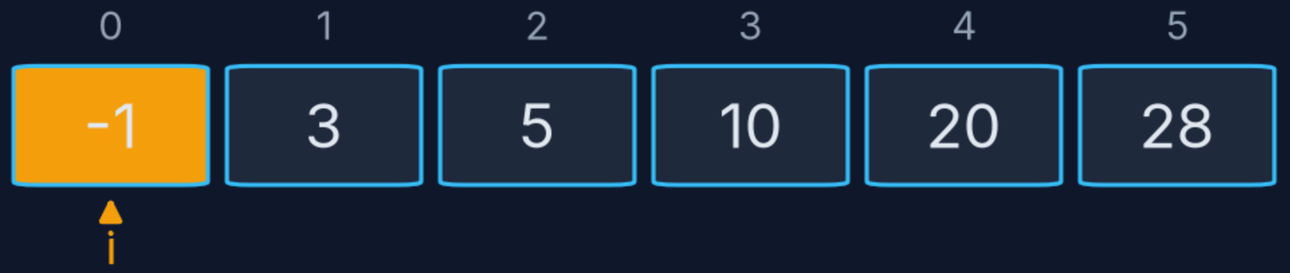
- Brute force: every pair → O(n·m)
- Sort both + two pointers → O(n log n + m log m)
- Moving the larger value can only widen the gap

Approach

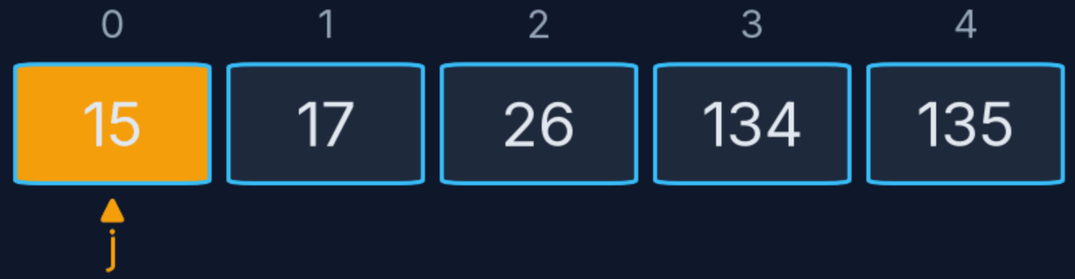
- 1. Sort both
- 2. Compare current values and record the gap
- 3. Advance the pointer at the smaller value
- 4. Stop early on zero

Worked example

A sorted



B sorted

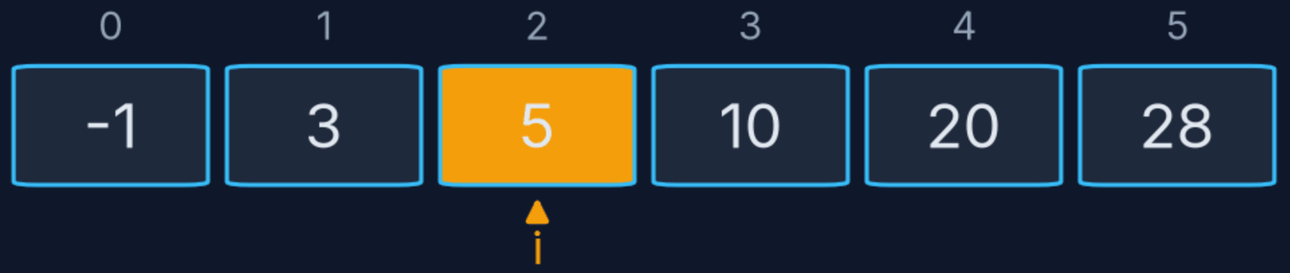


best = -1, 15 (16)

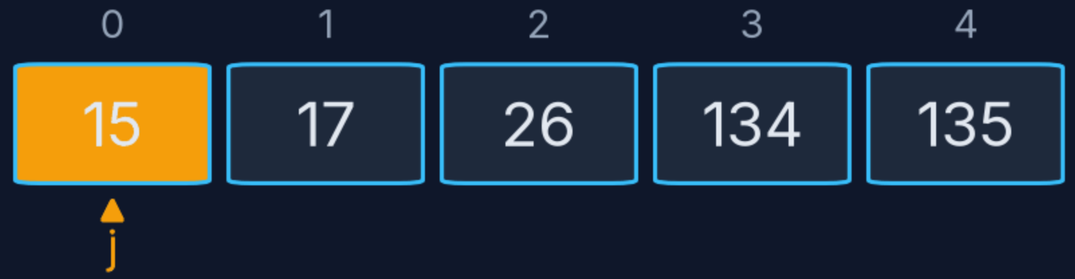
diff 16; move the smaller side

Worked example

A sorted



B sorted

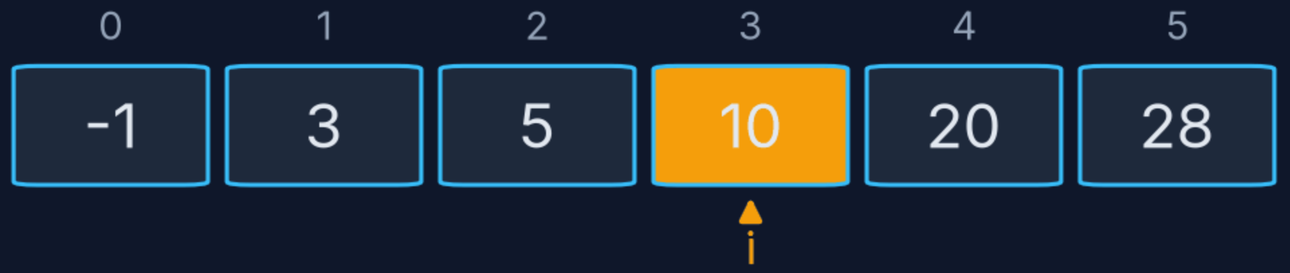



```
|5 - 15| = 10
best = 5, 15 (10)
```

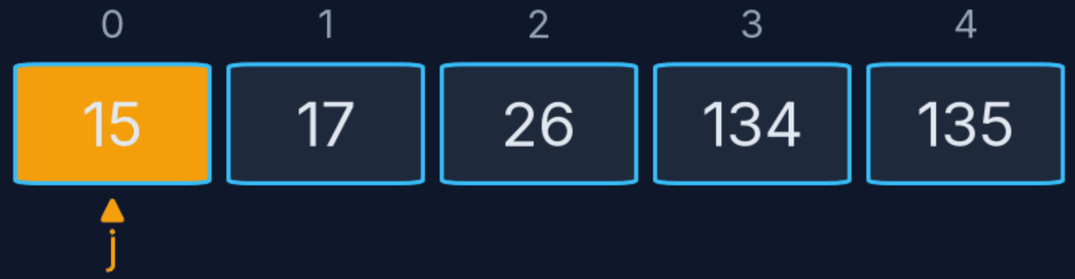
diff 10; move the smaller side

Worked example

A sorted



B sorted



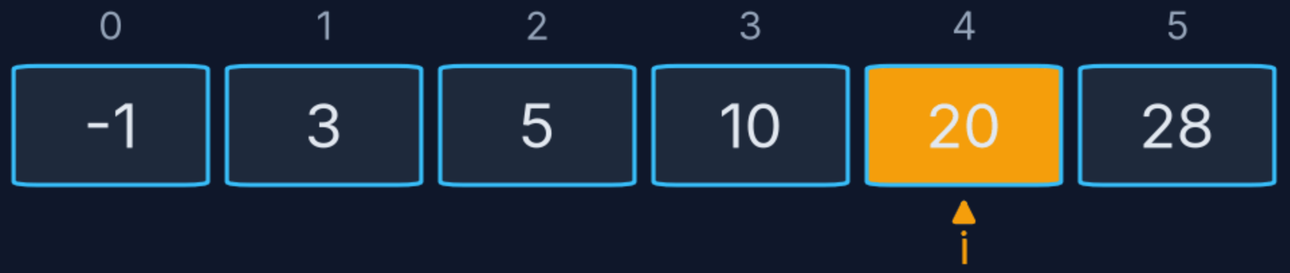
`|10 − 15| = 5`

`best = 10, 15 (5)`

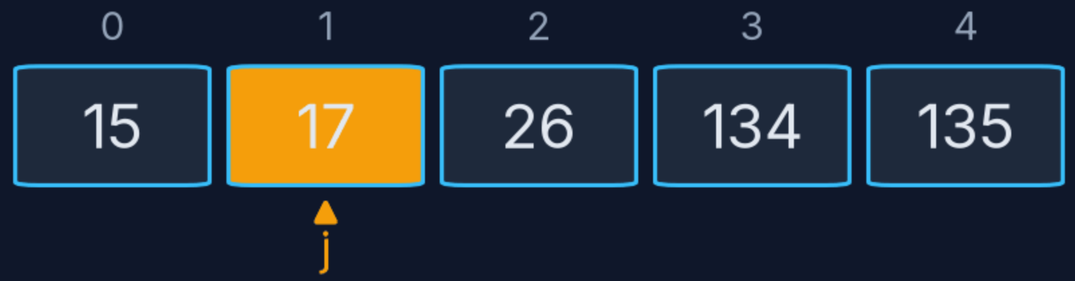
diff 5; move the smaller side

Worked example

A sorted



B sorted



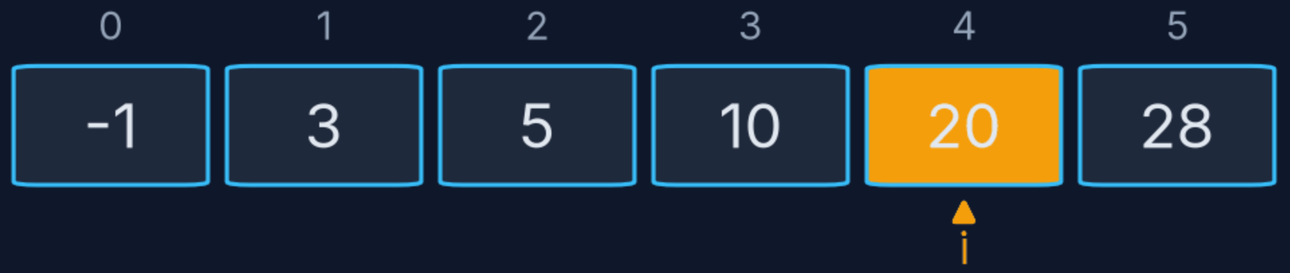
$|20 - 17| = 3$

best = 20, 17 (3)

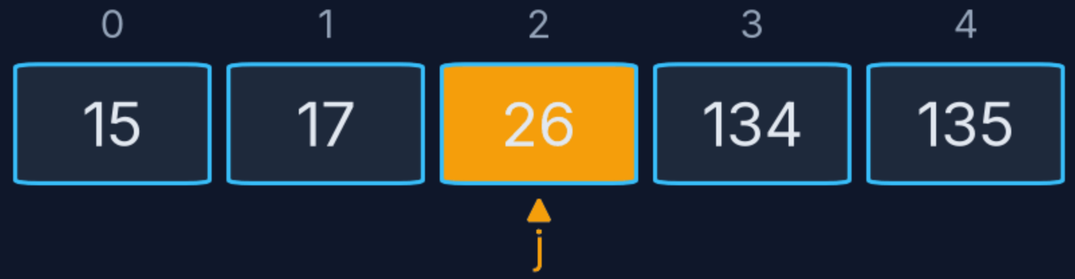
diff 3; move the smaller side

Worked example

A sorted



B sorted



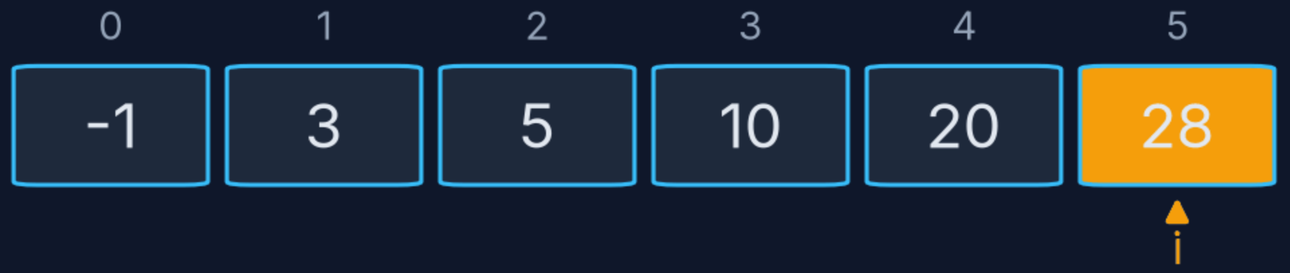
best = 20, 17 (3)

|20 − 26| = 6

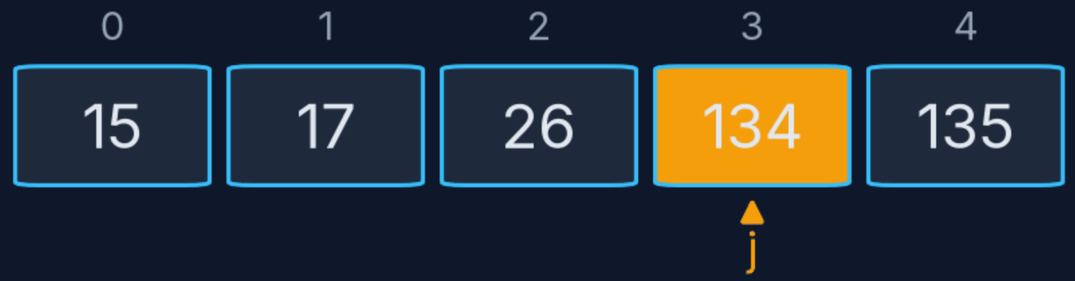
diff 6; move the smaller side

Worked example

A sorted



B sorted



best = 28 | 26 (2)

|28 − 134| = 106

diff 106; move the smaller side

Java solution

```java
public class SmallestDifference {
    public static int[] smallestDifference(int[] a, int[] b) {
        Arrays.sort(a);
        Arrays.sort(b);
        int i = 0, j = 0;
        long best = Long.MAX_VALUE;
        int[] pair = new int[2];
        while (i < a.length && j < b.length) {
            long d = Math.abs((long) a[i] - b[j]);
            if (d < best) { best = d; pair = new int[]{a[i], b[j]}; }
            if (a[i] < b[j]) i++; else if (a[i] > b[j]) j++; else break;
        }
        return pair;
    }
}
```

Complexity

- Time: O(n log n + m log m)
- Space: O(1)

Edge cases & pitfalls

- Equal values → difference 0, stop early
- Overflow on subtraction (use long)
- Negative numbers

Interview tips & follow-ups

- Explain why advancing the smaller is safe
- Follow-up: closest pair across 3 arrays

Recap

- Sort both arrays, then walk them together, always advancing the smaller value to close the gap.
- Complexity: O(n log n + m log m)